

WEBRUM-08: Dashboards and Alerting

Series: WEBRUM | **Notebook:** 8 of 8 | **Created:** March 2026 | **Last Updated:** 04/04/2026

Overview

An effective RUM monitoring strategy requires both visibility (dashboards) and proactive detection (alerting). Dashboards provide at-a-glance health views for different audiences — executives care about business impact and Apdex trends, while operations teams need real-time error rates and performance breakdowns.

This notebook covers building executive and operational RUM dashboards, calculating Apdex scores via DQL, creating RUM-based alerts for error rate spikes and performance degradation, and combining RUM with synthetic monitoring for a complete UX picture.

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Prerequisites

Requirement	Details
Dynatrace Environment	SaaS with Grail enabled
RUM Enabled	Web applications with at least 7 days of data
Permissions	<code>storage:events:read</code> , <code>storage:metrics:read</code> , dashboard create permissions
Previous Notebooks	All prior WEBRUM notebooks (01-07)

1. Executive RUM Dashboard KPIs

Executive dashboards should be simple, business-focused, and actionable. Key metrics:

KPI	What It Tells Executives	Target
Apdex score	Overall user satisfaction (0-1)	> 0.85
Session count	Traffic volume and trends	Depends on business
Error rate %	Percentage of sessions with errors	< 2%
Bounce rate %	Users leaving after one page	< 40%
CWV pass rate	% of page loads meeting all 3 CWV	> 75%
Avg session duration	User engagement level	Varies by app type

```
// Executive KPI summary – single-query dashboard tile
fetch user.sessions, from:-24h
| filter userType == "REAL_USER"
| summarize
    total_sessions = count(),
    error_sessions = countIf(totalErrorCount > 0),
    bounce_sessions = countIf(userActionCount == 1),
    avg_actions = avg(userActionCount),
    avg_duration_min = avg(toDouble(duration) / 60000000000.0),
    by:{application}
| fieldsAdd error_rate_pct = round(toDouble(error_sessions) /
toDouble(total_sessions) * 100.0, decimals: 1),
    bounce_rate_pct = round(toDouble(bounce_sessions) /
toDouble(total_sessions) * 100.0, decimals: 1),
    avg_duration_min = round(avg_duration_min, decimals: 1)
| sort total_sessions desc
```

```
// Session volume trend – 7-day daily session counts
fetch user.sessions, from:-7d
| filter userType == "REAL_USER"
| makeTimeseries session_count = count(), interval:1d, by:{application}
```

2. Apdex Score Calculation

The **Application Performance Index (Apdex)** is an open standard for measuring user satisfaction. It classifies every user action into three buckets based on a threshold T:

Classification	Duration	Description
Satisfied	$\leq T$	Action completed within acceptable time
Tolerating	$T < \text{duration} \leq 4T$	Action was slow but user waited
Frustrated	$> 4T$	Action was too slow or failed

Apdex formula:

```
Apdex = (Satisfied + (Tolerating / 2)) / Total
```

The threshold T is configurable per application. Common defaults:

Action Type	Default T
Page load	3 seconds
XHR action	2.5 seconds
Route change	2.5 seconds

```
// Apex calculation with T = 3 seconds for page loads
fetch user.events, from:-24h
| filter action.type == "Load"
| fieldsAdd duration_sec = toDouble(duration) / 1000000000.0
| summarize total = count(),
    satisfied = countIf(duration_sec <= 3),
    tolerating = countIf(duration_sec > 3 and duration_sec <= 12),
    frustrated = countIf(duration_sec > 12),
    by:{application}
| fieldsAdd apdex = round((toDouble(satisfied) + toDouble(tolerating) / 2.0)
/ toDouble(total), decimals: 3)
| sort apdex asc
```

```
// Apex trend over 7 days – daily Apex score
fetch user.events, from:-7d
| filter action.type == "Load"
| fieldsAdd duration_sec = toDouble(duration) / 1000000000.0
| fieldsAdd is_satisfied = if(duration_sec <= 3, 1.0, else: 0.0),
    is_tolerating = if(duration_sec > 3 and duration_sec <= 12, 0.5,
else: 0.0)
| makeTimeseries
    total_actions = count(),
    satisfied_score = sum(is_satisfied),
    tolerating_score = sum(is_tolerating),
    interval:1d
```

```
// Apex by page – which pages have the worst user satisfaction?
fetch user.events, from:-24h
| filter action.type == "Load"
| fieldsAdd duration_sec = toDouble(duration) / 1000000000.0
| summarize total = count(),
    satisfied = countIf(duration_sec <= 3),
    tolerating = countIf(duration_sec > 3 and duration_sec <= 12),
    by:{action.name}
| filter total > 20
| fieldsAdd apdex = round((toDouble(satisfied) + toDouble(tolerating) / 2.0)
/ toDouble(total), decimals: 3)
| sort apdex asc
| limit 10
```

Interpreting Apdex Scores

Apdex Range	Rating	Action
0.94 - 1.00	Excellent	No action needed
0.85 - 0.93	Good	Monitor for changes
0.70 - 0.84	Fair	Investigate slow pages
0.50 - 0.69	Poor	Prioritize optimization
< 0.50	Unacceptable	Immediate attention required

3. Operational RUM Dashboard

Operations teams need real-time visibility into errors, performance anomalies, and traffic patterns.

```
// Real-time error rate – errors per 15 minutes over the last 6 hours
fetch user.events, from:-6h
| filter type == "Error"
| makeTimeseries error_count = count(), interval:15m, by:{error.type}
```

```
// Active error summary – current top errors in the last hour
fetch user.events, from:-1h
| filter type == "Error"
| summarize error_count = count(),
    affected_sessions = countDistinct(sessionId),
    by:{error.message, error.type, application}
| sort affected_sessions desc
| limit 10
```

```
// Performance SLA – percentage of page loads under 3 seconds
fetch user.events, from:-1h
| filter action.type == "Load"
| fieldsAdd duration_sec = toDouble(duration) / 1000000000.0
| summarize total = count(),
    under_3s = countIf(duration_sec <= 3),
    by:{application}
| fieldsAdd sla_pct = round(toDouble(under_3s) / toDouble(total) * 100.0,
    decimals: 1)
| sort sla_pct asc
```

4. RUM Alerting Strategies

Effective RUM alerting requires the right balance — too sensitive triggers alert fatigue, too loose misses real issues.

Recommended Alert Categories

Category	What to Alert On	Threshold Example
Error rate spike	Sudden increase in JS error rate	> 5% of sessions with errors
Apdex drop	User satisfaction below threshold	Apdex < 0.7 for 15 minutes
Performance degradation	p75 page load exceeds baseline	p75 > 5s for 15 minutes
Traffic anomaly	Unexpected drop in session volume	< 50% of previous day's hourly average

Category	What to Alert On	Threshold Example
CWV regression	Core Web Vitals crossing into Poor	LCP p75 > 4s for 30 minutes

Alert Configuration via Metrics Events

Dynatrace metric events allow alerting on RUM metrics. Configure via:

Settings > Anomaly detection > Metric events > Add metric event

Setting	Recommended Value
Metric	RUM metric (e.g., <code>builtin:apps.web.action.apdex</code>)
Evaluation window	15 minutes (balances speed and noise)
Sliding window	5-minute intervals
Threshold type	Static or auto-adaptive baseline

5. Error Rate Alerts

The following queries provide the data foundation for error rate alerting. Use these patterns to detect when error rates exceed acceptable thresholds.

```
// Error rate per 15-minute window – alert threshold data
fetch user.sessions, from:-6h
| filter userType == "REAL_USER"
| fieldsAdd has_error = if(totalErrorCount > 0, 1.0, else: 0.0)
| makeTimeseries
  total_sessions = count(),
  error_sessions = sum(has_error),
  interval:15m,
  by:{application}
```

```
// New errors – errors first seen in the last getHour(potential deployment issue)
fetch user.events, from:-1h
| filter type == "Error"
| summarize first_seen = min(timestamp),
    error_count = count(),
    affected_sessions = countDistinct(sessionId),
    by:{error.message, application}
| sort first_seen desc
| limit 10
```

6. Performance Degradation Alerts

Detect when page load performance degrades beyond acceptable thresholds.

```
// p75 page load duration per 15-minute window – performance alert data
fetch user.events, from:-6h
| filter action.type == "Load"
| fieldsAdd duration_ms = toDouble(duration) / 1000000.0
| makeTimeseries p75_duration = percentile(duration_ms, 75), interval:15m,
by:{application}
```

```
// Apdex per 15-minute window – satisfaction alert data
fetch user.events, from:-6h
| filter action.type == "Load"
| fieldsAdd duration_sec = toDouble(duration) / 1000000000.0
| fieldsAdd apdex_score = if(duration_sec <= 3, 1.0,
    else: if(duration_sec <= 12, 0.5,
    else: 0.0))
| makeTimeseries avg_apdex = avg(apdex_score), interval:15m, by:{application}
```

Tip: For production alerting, use Dynatrace Intelligence anomaly detection rather than static thresholds. Dynatrace Intelligence automatically baselines normal behavior and detects deviations, reducing false positives from seasonal traffic patterns.

7. RUM + Synthetic Combined View

RUM and synthetic monitoring complement each other. Combining them provides a complete UX monitoring picture:

Aspect	RUM Covers	Synthetic Covers
Availability	Only when users visit	24/7 scheduled monitoring
Baseline	Real-world conditions (variable)	Clean-room conditions (stable)
Coverage	Pages users visit	All critical paths (scripted)
Error detection	Errors users hit	Errors in scripted flows
Geographic	Where users are	From configured locations

Side-by-Side Comparison

Use `append` to compare RUM and synthetic performance for the same application:

```
// RUM session summary – real user experience
fetch user.sessions, from:-24h
| filter userType == "REAL_USER"
| summarize rum_sessions = count(),
  rum_error_sessions = countIf(totalErrorCount > 0),
  rum_avg_actions = avg(userActionCount),
  by:{application}
| fieldsAdd monitoring_type = "RUM",
  error_rate_pct = round(toDouble(rum_error_sessions) /
toDouble(rum_sessions) * 100.0, decimals: 1)
```

```
// Synthetic vs RUM – compare session types for the same app
fetch user.sessions, from:-24h
| summarize session_count = count(),
  avg_duration_sec = avg(toDouble(duration) / 1000000000.0),
  error_sessions = countIf(totalErrorCount > 0),
  by:{application, userType}
| filter userType == "REAL_USER" or userType == "SYNTHETIC"
| fieldsAdd error_rate_pct = round(toDouble(error_sessions) /
toDouble(session_count) * 100.0, decimals: 1)
| sort application asc, userType asc
```

8. Summary and Series Recap

In this notebook, we covered:

- **Executive KPIs** — Session count, error rate, bounce rate, Apdex
- **Apdex calculation** — Satisfied/tolerating/frustrated classification via DQL
- **Operational dashboard** — Real-time error rates, SLA tracking, active errors
- **Alerting strategies** — Error rate spikes, Apdex drops, performance degradation
- **RUM + Synthetic** — Combining both monitoring types for complete coverage

WEBRUM Series Recap

Notebook	Topic	Key Takeaway
01	RUM Fundamentals	JavaScript agent, data model, basic queries
02	SPA Instrumentation	Route changes, framework tips, XHR monitoring
03	Core Web Vitals	LCP, INP, CLS measurement and scoring
04	Session Analysis	Segmentation, journeys, conversions, bounce rates
05	Error Analysis	Error impact, rage clicks, error-session correlation
06	Performance Analysis	Waterfall, TTFB, slow pages, geographic performance
07	Session Replay	Privacy, masking, finding and correlating replays
08	Dashboards & Alerting	Apdex, KPI dashboards, alerting strategies

References

- [Apdex Standard](#)
 - [Dynatrace RUM Dashboards](#)
 - [Metric Events for Alerting](#)
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